

QUICKMEDS-AN ONLINE PHARMACY

**Prof. S. K. Kharkate^{*1}, Tanmay M. Warthe^{*2}, Lobhas V. Gawande^{*3}, Shyam P. Murodiya^{*4},
Fariha S. Sheikh^{*5}, Rhushabh V. Gumte^{*6}**

^{*1}HOD Of Computer Technology, Government Polytechnic Bramhapuri, Maharashtra, India.

^{*2,3,4,5,6}Student Of Computer Technology, Government Polytechnic Bramhapuri, Maharashtra, India.

ABSTRACT

A pharmacy management system is primarily designed to facilitate the efficient management of medicine supplies required by a hospital. This system plays a crucial role in ensuring that necessary medicines are readily available, thereby supporting timely and effective treatment for hospital patients. It utilizes information technology and databases to serve as a centralized repository for essential data, helping streamline hospital operations and improve overall management. To build a reliable and efficient system, various software tools and technologies are integrated.

QuickMeds is an advanced online pharmacy platform developed with the goal of simplifying the process of ordering medicines and healthcare products. It provides users with an easy-to-use interface to manage prescriptions, track orders, and receive home delivery of essential items. By bridging the gap between pharmacies and customers, QuickMeds enhances convenience and ensures accessibility. The system is designed to serve both patients and healthcare professionals, enabling smooth interaction between users and pharmacists for a better healthcare experience [5].

Keywords: E-Pharmacy, Online Platform, Medicine Purchase, Prescription Management, Healthcare System.

I. INTRODUCTION

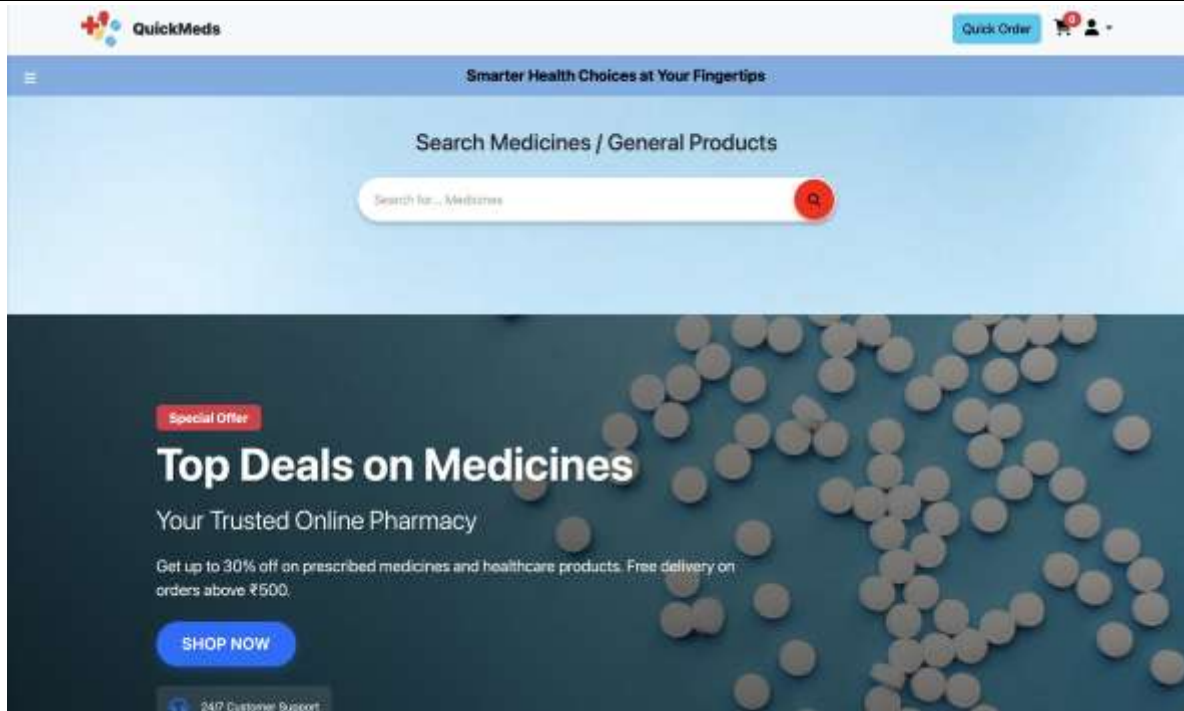
In recent years, online pharmacies have gained significant popularity due to the convenience they provide. Customers are no longer required to visit physical stores to purchase medications, as online platforms allow them to order from the comfort of their homes. This advancement has improved access to healthcare, particularly for individuals with mobility issues or those living in remote areas. Despite the increasing adoption of e-pharmacy services, several challenges persist. These include limitations in user experience, concerns over prescription verification, product authenticity, pricing inconsistencies, and lack of access to affordable generic alternatives. Such issues impact the overall effectiveness and safety of online pharmacy platforms. This project focuses on addressing the major drawbacks of existing online pharmacy systems by proposing QuickMeds—a robust and user-friendly platform. QuickMeds enables users to upload prescriptions, search for healthcare products, compare prices, and place orders with ease. It also allows pharmacies to manage inventory, track orders, and provide support services. The goal is to enhance medicine accessibility and streamline pharmacy operations while encouraging fair pricing and promotion of generic medicines [3].

1.2 Aim of the Research:

The aim of this project is to redefine the sale of pharmaceutical and lifestyle-essential products using an online platform. It also strives to promote pricing equality among different brands offering identical pharmaceutical drugs and to encourage the use of affordable generic alternatives.

II. METHODOLOGY

The Quickmeds application is designed to offer a seamless and efficient online medicine ordering experience. The system follows a modular development approach, focusing on core functional areas required for managing medical products, users, and orders [5].



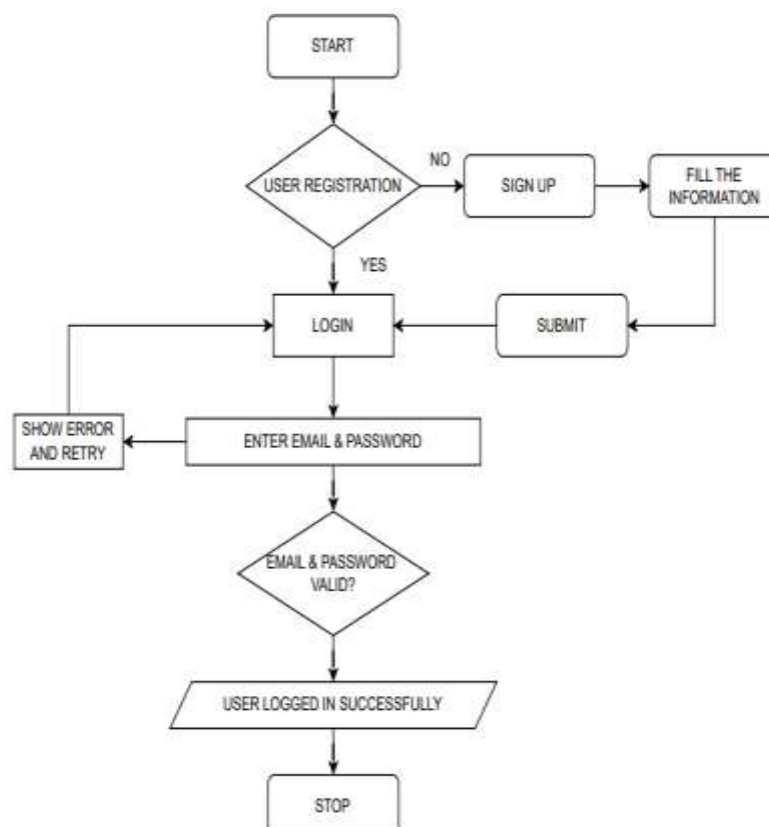
The application is divided into two primary roles: Admin and Customer, with several interconnected sub-modules as described below.

Sub-Modules of Quickmeds :

1. Module 1: User Management/Login Flowchart

This module handles the User Registration and Login for customers and admins, Authentication and Authorization and Profile Management (view/update personal details).

Security and privacy are ensured through proper validation and encrypted storage of sensitive data.



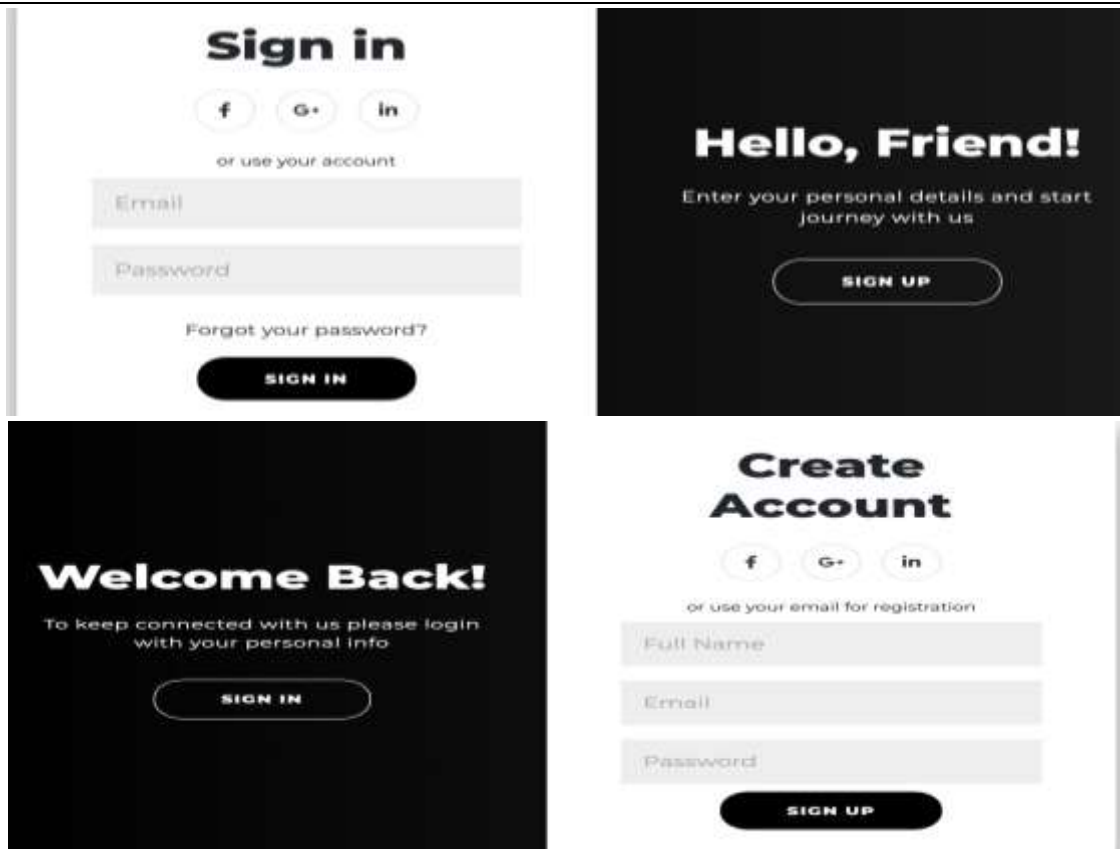
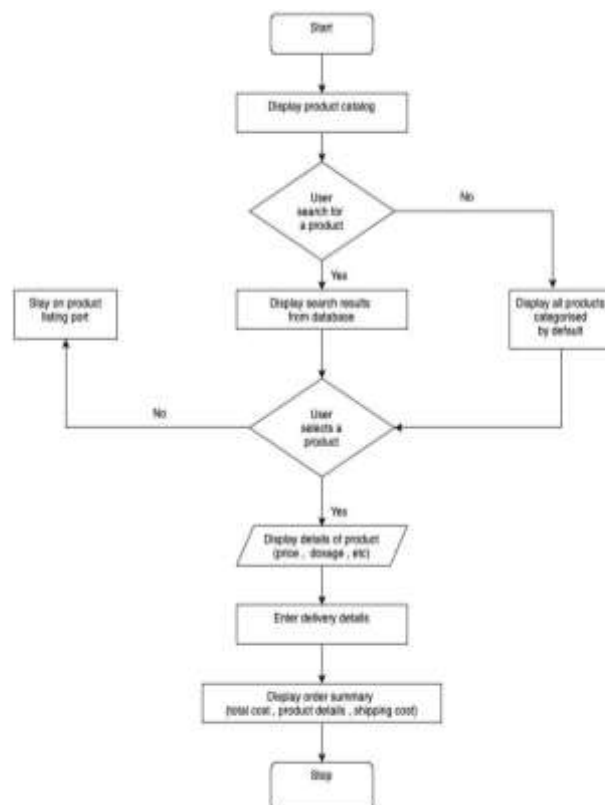


Fig. 2.1 User Management/Login Flowchart

2. Module 2: Product Search and Catalogue Flowchart

This module maintains all information related to available medicines like Adding and updating product details (by Admin), Viewing/searching medicines (by Customers) and Filtering by categories, brands, and price.

Medicines include details like name, description, price, manufacturer, stock, and expiry date.



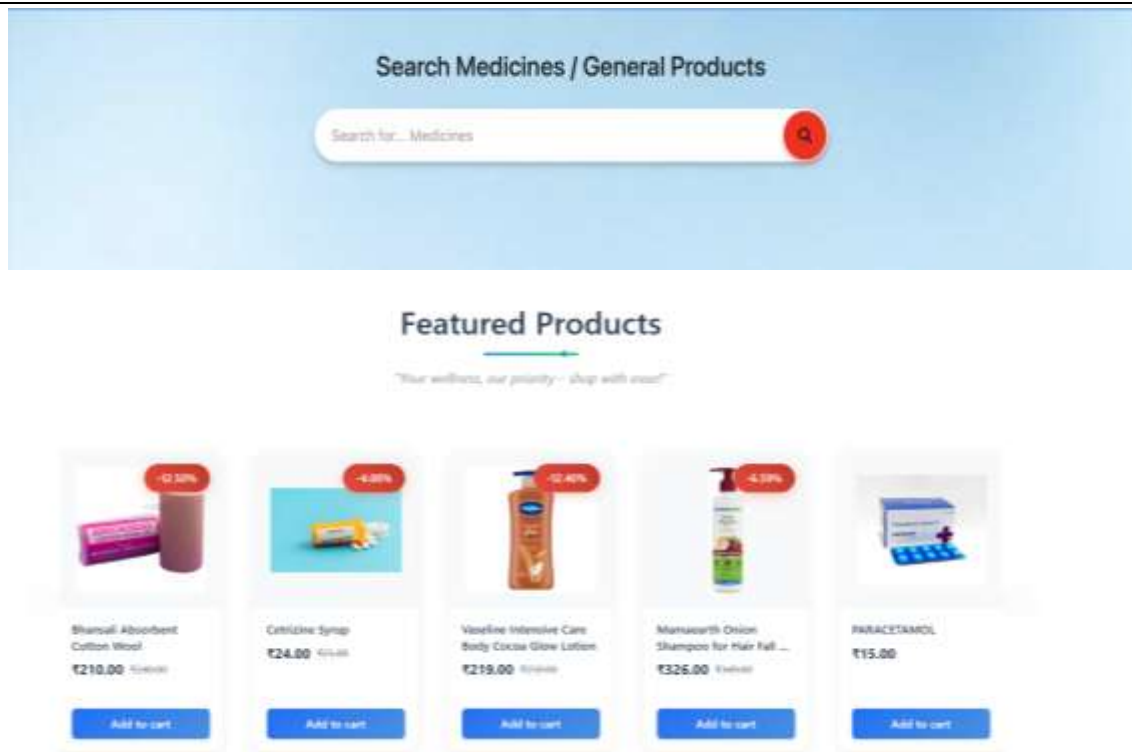


Fig. 2.2 Product Search and Catalogue Flowchart

3. Module 3: Add to Cart and Checkout Flowchart

Customers can Add or remove items from the cart, View cart summary with quantities and total cost, Proceed to checkout by confirming delivery address.

The module provides a smooth flow from product selection to order placement.



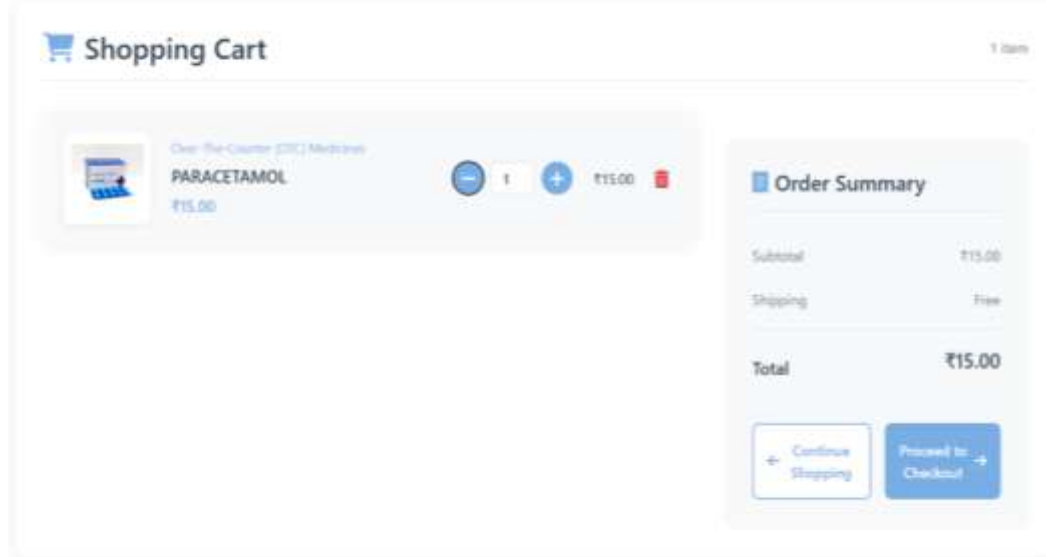


Fig. 2.3 Add to Cart and Checkout Flowchart

4. Module 4: Order Tracking Flowchart

Both customers and admins interact with this module. Here, the Customers can view order status (e.g., Pending, Confirmed, Shipped) and Admins can manage and update order details.

Order history is maintained for transparency and tracking.

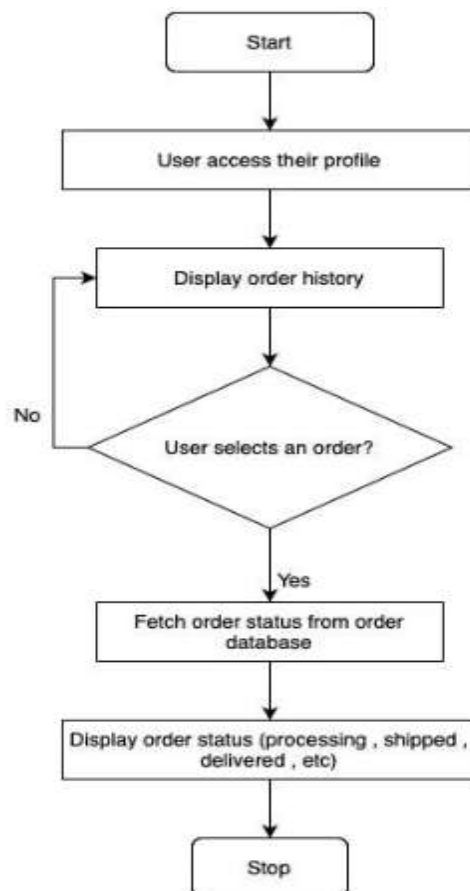


Fig. 2.4 Order Tracking Flowchart

5. Module 5: Payment Method Flowchart

This module integrates various payment gateways to offer Secure transaction processing, Multiple payment options like UPI and COD.

Sensitive data is securely transmitted and handled following best practices.

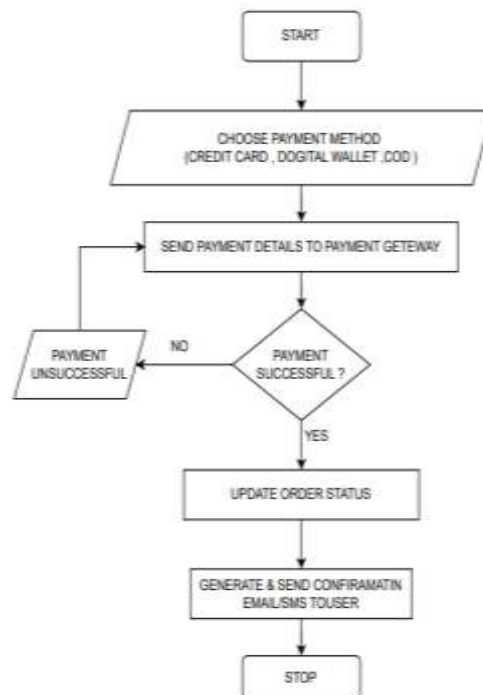
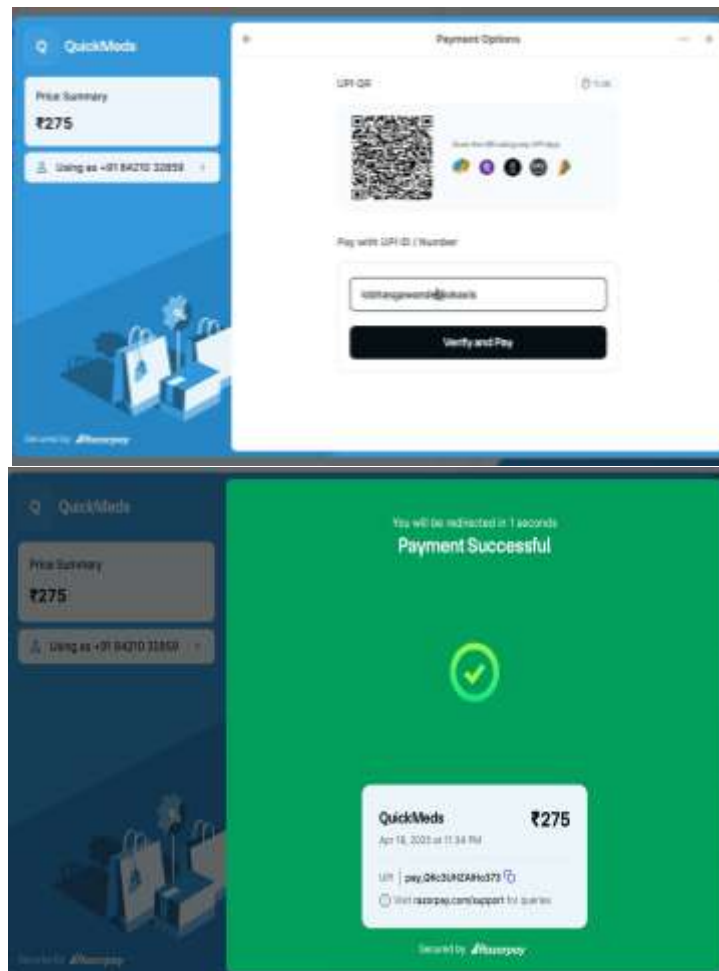


Fig. 2.5 Payment Method Flowchart



6. Module 6: Admin Inventory Management Flowchart

Admins have full control over the platform to Manage user accounts, Oversee product listings and orders, Monitor payments and deliveries and Generate reports and insights for decision-making.

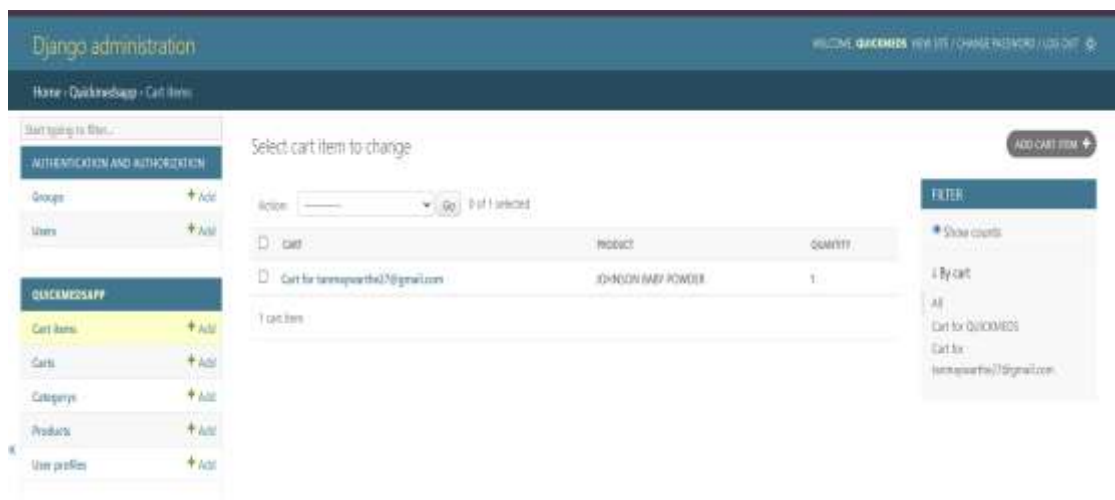
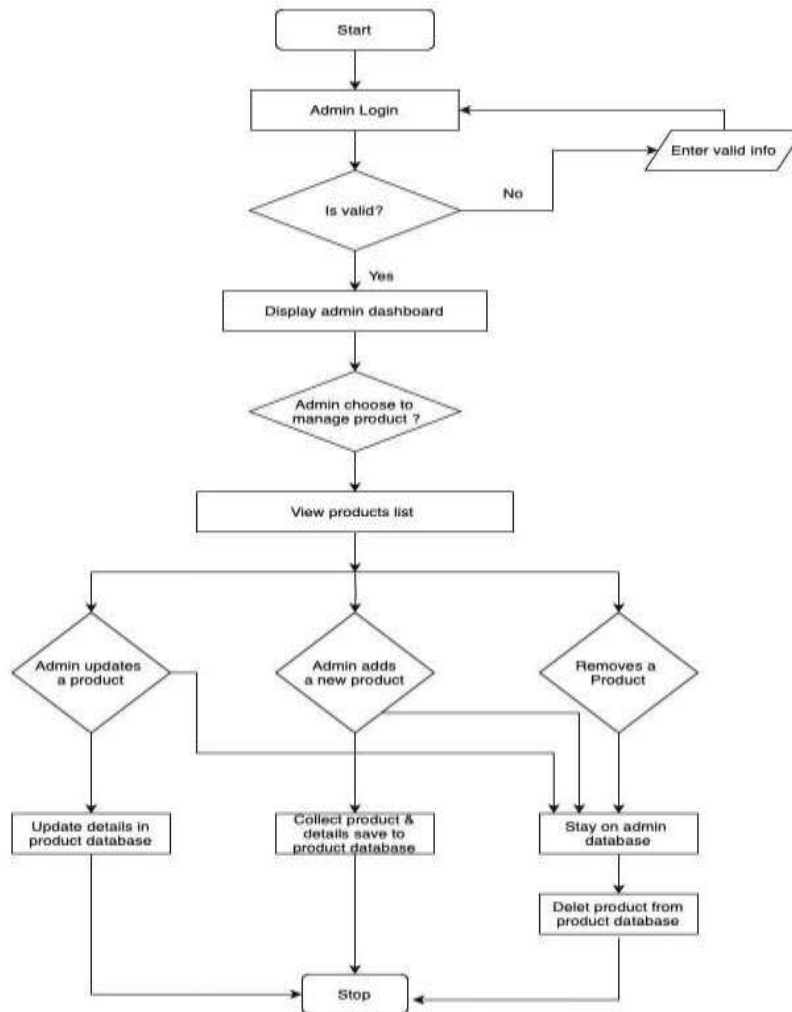


Fig. 2.6 Admin Inventory Management Flowchart

III. RESULTS AND DISCUSSION

The system was developed with several important modules, each helping to improve how the platform works for both users and admins. The results show that the system runs smoothly, is easy to use, and keeps user data safe.

1. Registration and Login: The registration module made it easy for new users to create their accounts. The login system helped protect the platform by allowing only authorized users, especially admins, to access certain features. This helped keep user data and system functions secure.

2. Product Management: Admins could easily add, update, or remove product listings using the product management modules. This kept the product information well-organized and up to date, helping customers to find what they needed without confusion.

3. Order and Payment: Placing orders through the system was simple and quick. Users could view their order history, which was helpful for keeping track of past purchases. The payment module supported multiple payment options, making checkout flexible and secure for customers.

4. System Usability: The interface was clean and easy to understand, which made the system user-friendly. Each module was connected in a smooth workflow, making tasks like browsing, buying, and managing products feel seamless.

5. Development Experience: Some challenges came up, like setting up secure payment options and making sure the data updated instantly. These were handled with proper coding techniques and the use of reliable tools. Overall, the development process helped improve technical skills and understanding of modular design [6].

IV. CONCLUSION

QuickMeds is designed to simplify and enhance the experience of ordering and receiving medications through a secure and efficient online platform. By allowing users to upload prescriptions, search for a wide range of products, and track their orders in real-time, the system reduces the dependency on physical pharmacy visits and makes healthcare more accessible, especially for the elderly and people in remote areas.

The platform not only benefits end-users but also empowers pharmacists by offering tools to manage inventory, process orders, and communicate with customers, all from a centralized interface. This leads to improved operational efficiency, better stock handling, and faster fulfillment of medical needs.

By integrating strong security measures and user-friendly design, QuickMeds addresses major concerns like data privacy, product verification, and delivery delays. It ensures that only authenticated and quality-assured medicines are delivered through verified vendors.

Ultimately, QuickMeds bridges the gap between pharmacies and patients, bringing essential healthcare services directly to users' doorsteps and promoting the safe, timely, and affordable consumption of medications. It represents a step forward in modernizing the pharmaceutical sector with technology-driven solutions that prioritize safety, convenience, and reliability [2].

V. REFERENCES

- [1] Eysenbach, G. (2001). Online pharmacy: Legal, regulatory, and economic aspects. Journal of Medical Internet
- [2] Research, vol.3.
- [3] Ashwin Kumar Chaturvedi , Umesh Kumar Singh, Amrish Kumar "ONLINE PHARMACY: AN E-STRATEGY FOR MEDICATION,IJPFR, April-June 2011;
- [4] Hewa Majeed Zangana , Design an Information Management System for a Pharmacy,Vol. 7, Issue 10, October 2018
- [5] Wirachchaya Chanpuypetch & Duangpun Kritchanhai (2017): International Journal of Healthcare Management
- [6] Renjithkumar Surendran Pillai , "Quickmeds" An Online Market for Medicine, 2 December 2024